

Ahmed Elsayed Taha

AI Engineer | Machine Learning Engineer | NLP & LLM Engineer

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PROFILE SUMMARY

AI Engineer with hands-on experience designing and deploying LLM-based applications, including retrieval-augmented generation (RAG) pipelines, multi-agent systems, and fine-tuned language models using QLoRA and LoRA. Proficient in Python, Hugging Face Transformers, LangChain, and LangGraph, with practical experience moving AI prototypes to production using FastAPI, Docker, and vector databases such as FAISS. Solid foundation in machine learning and deep learning from a B.Sc. in Artificial Intelligence & Data Science, with a demonstrated track record of applying generative AI and NLP to solve real-world problems including financial reasoning QA, Arabic dialect adaptation, and automated multi-agent research systems.

TECHNICAL SKILLS

- Programming Languages: Python, C++, JavaScript, SQL
- Machine Learning: TensorFlow, Keras, Scikit-learn, XGBoost, Feature Engineering, Model Evaluation
- Deep Learning: Neural Networks, CNNs, Transfer Learning, PyTorch-compatible workflows
- LLMs & Generative AI: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning (LoRA, QLoRA, PEFT), Hugging Face Transformers, TRL, BitsAndBytes, Vector Databases (FAISS), Embeddings
- NLP: Text Preprocessing, Tokenization, Instruction Dataset Curation, Dialect/Style Adaptation, Conversational AI
- Frameworks & Tools: Hugging Face Datasets, Streamlit
- Cloud & Deployment: AWS S3, Docker, FastAPI, Temporal Workflows, Model Deployment
- Databases: FAISS (Vector Database), SQL
- Version Control & Collaboration: Git, GitHub
- Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA)

WORK EXPERIENCE

Data Analysis Intern — DEY, Creativa Innovation Hubs

01/2026 – 03/2026

- Performed exploratory data analysis (EDA) and feature engineering on real-world datasets to support data-driven decision-making.

AI/ML Training Program — National Telecommunication Institute (NTI)

07/2023 – 09/2023

- Completed intensive summer training in artificial intelligence and machine learning, covering data preprocessing, ML model development, and practical AI applications.

PROJECTS

Financial Analyst AI Assistant — LLM Fine-Tuning Project

Technologies: Qwen2.5-1.5B-Instruct, QLoRA, PEFT, Hugging Face Transformers, BitsAndBytes

- Problem: Needed a conversational assistant capable of answering financial questions grounded in company reports.
- Solution: Fine-tuned Qwen2.5-1.5B-Instruct using QLoRA (LoRA plus 4-bit quantization) on 8,000+ financial reasoning QA samples, preparing instruction datasets with Hugging Face Datasets and Qwen chat templates for supervised fine-tuning.
- Result: Delivered a memory-efficient, domain-adapted LLM assistant optimized for consumer-grade hardware using PEFT and BitsAndBytes.

Egyptian Arabic Dialect Style Adapter

Technologies: Qwen, QLoRA, LoRA, PEFT, Hugging Face Transformers, TRL

- Problem: General-purpose LLMs lacked a natural conversational style in Egyptian Arabic dialect.

- Solution: Curated and cleaned approximately 57,000 dialogue examples from business, e-commerce, and conversational sources into SFT-ready instruction format, then fine-tuned Qwen using QLoRA (4-bit NF4 quantization) with gradient accumulation via Hugging Face Transformers and TRL.
- Result: Evaluated dialect adaptation quality and published the resulting LoRA adapter and dataset artifacts publicly on Hugging Face.

Deep Research Agent — Multi-Agent AI System

Technologies: LangGraph, LangChain, Groq LLMs, Tavily API

- Problem: Manual research and report writing is slow and inconsistent across sources.
- Solution: Designed a four-agent research system (Planner, Researcher, Writer, Critic) with LangGraph and LangChain, integrating Groq LLMs and the Tavily API to automate multi-step research.
- Result: Automated generation of structured, citation-backed PDF research reports.

YouTube Playlist RAG System

Technologies: LangChain, FAISS, Hugging Face Embeddings, Flask

- Problem: Long-form YouTube playlist content is difficult to search and query for specific information.
- Solution: Built a RAG pipeline that extracts YouTube transcripts, generates embeddings, and indexes them in a FAISS vector database.
- Result: Enabled context-aware, source-grounded answers to user queries retrieved directly from playlist content.

AI Contract Review System

Technologies: Temporal, FastAPI, OpenRouter, AWS S3

- Problem: Reviewing multiple contracts for risk and key terms is time-consuming and inconsistent when done manually.
- Solution: Built an AI-powered contract review system using Temporal Parent/Child workflows to analyze multiple contracts in parallel, with LLM-based summarization and cross-contract risk analysis; developed REST APIs with FastAPI and fault-tolerant orchestration (retries, heartbeats, S3-based PDF processing).
- Result: Enabled human-in-the-loop review via Queries, Signals, and Updates for reliable, auditable contract analysis.

EDUCATION

B.Sc. in Artificial Intelligence & Data Science — Benha University (BFCAI)

2020 – 2024

GPA: 3.41 / 4.00

- Graduation Project (Grade: A+): Alzheimer's Detection with Generative AI — developed a deep learning model to detect Alzheimer's disease from 33,000+ MRI images (preprocessing and augmentation), with a GPT-2 chatbot to explain results, deployed via Flask and Firebase.

CERTIFICATIONS

- Generative AI with Large Language Models — designed and evaluated RAG architectures, analyzing cost, latency, and quality trade-offs
- Machine Learning Scientist in Python — DataCamp (2025)
- Google Data Analytics Professional Certificate
- AWS Academy Data Engineering

LANGUAGES

- Arabic (Native)
- English (Professional Working Proficiency)